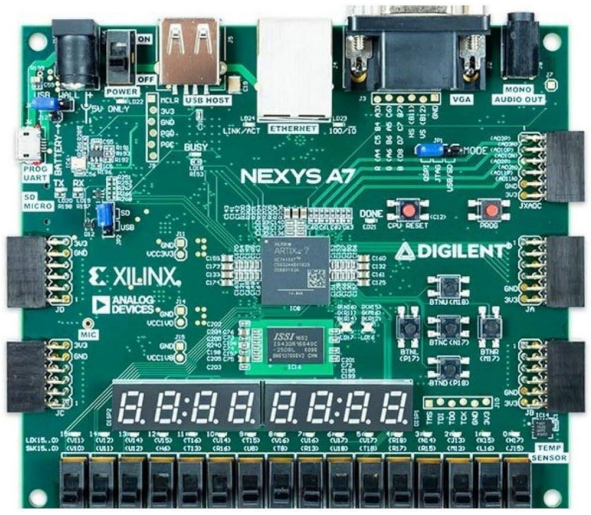
Accelerometer-Driven Display Rotator and Tilt Level Meter

Objectives

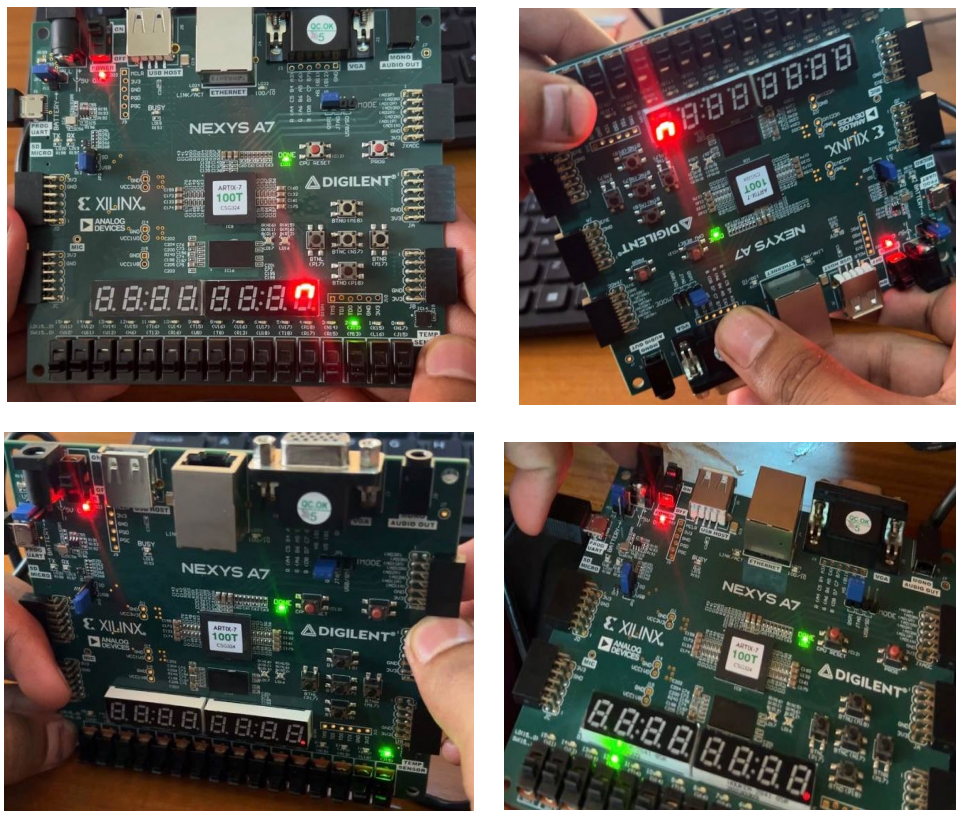
- ❑ Measure **tilt and rotation** using the **accelerometer** on the NEXYS A7 FPGA .
- ❑ Compute **roll and pitch angles.**
- ❑ Display tilt using **LEDs** and rotation using **7-segment** display.
- ❑ Ensure **stable real-time** operation.

Key Features

- ✓ **Real-Time Orientation Detection** using onboard accelerometer sensor and **FPGA-Based Processing**.
- ✓ **Roll Angle Visualization** through rotating the character on the 7-segment display.
- ✓ **Pitch Angle Level Meter** using multiple LEDs to indicate tilt magnitude.



FPGA Board



FPGA Implementation

YouTube Link

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